RACE Regular Analysis Through Continuous Exercise

Statistics: RACE-01

Single Correct Type Questions:

1. The mean of discrete observations $y_1, y_2, \ldots\ldots, y_n$ is given by

(A) $\frac{\sum_{i=1}^{n} y_i}{n}$ (B) $\frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} i}$ (C) $\frac{\sum_{i=1}^{n} y_i f_i}{n}$ (D) $\frac{\sum_{i=1}^{n} y_i f_i}{\sum_{i=1}^{n} f_i}$

MSI152

2. If the mean of the numbers $27+x, 31+x, 89+x, 107+x, 156+x$ is 82, then the mean of $130+x, 126+x, 68+x, 50+x, 1+x$ is

(A) 75 (B) 157 (C) 82 (D) 80

MSI153

3. The number of observations in a group is 40. If the average of first 10 is 4.5 and that of the remaining 30 is 3.5, then the average of the whole group is

(A) $\frac{1}{5}$ (B) $\frac{15}{4}$ (C) 4 (D) 8

MSI154

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4. Mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50 respectively. The correct mean is

(A) 48 (B) $82\frac{1}{2}$ (C) 50 (D) 80

MSI155

5. Consider the frequency distribution of the given numbers

Value:	1	2	3	4
Frequency:	5	4	6	f

If the mean is known to be 3, then the value of f is

(A) 3 (B) 7 (C) 10 (D) 14

MSI156

6. Find the mean wage from the following data:

Wage (in Rs)	800	820	860	900	920	980	1000
No. of workers:	7	14	19	25	20	10	5

(A) 890.2 (B) 899.2 (C) 891.2 (D) 892.2

MSI157

7. The weighted mean of first n natural numbers whose weights are equal to the squares of corresponding numbers is

(A) $\frac{n+1}{2}$ (B) $\frac{3n(n+1)}{2(2n+1)}$ (C) $\frac{(n+1)(2n+1)}{6}$ (D) $\frac{n(n+1)}{2}$

MSI158

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8. If the arithmetic means of the numbers $x_1, x_2, x_3, \ldots\ldots, x_n$ is $\bar{x}$, then the arithmetic means of numbers $ax_1 + b, ax_2 + b, ax_3 + b, \ldots\ldots\ldots, ax_n + b$, where a, b are two constants would be

(A) $\bar{x}$ (B) $na\bar{x} + nb$ (C) $a\bar{x}$ (D) $a\bar{x} + b$

MSI159

9. In a class of 100 students there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class are 72, then what are the average marks of the girls

(A) 73 (B) 65 (C) 68 (D) 74

MSI160

10. The mean weight per student in a group of seven students is 55 kg. If the individual weights of 6 students are 52, 58, 55, 53, 56 and 54, then weight of the seventh student is

(A) 55 kg (B) 60 kg (C) 57 kg (D) 50 kg

MSI161

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	B	C	D	C	B	D	B	C

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Regular Analysis Through Continuous Exercise

Statistics: RACE-02

Single Correct Type Questions:

1. If a variable takes the discrete values $\alpha+4,\alpha-\frac{7}{2},\alpha-\frac{5}{2},\alpha-3,\alpha-2,\alpha+\frac{1}{2},\alpha-\frac{1}{2},\alpha+5(\alpha>0)$, then the median is

(A) $\alpha-\frac{5}{4}$ (B) $\alpha-\frac{1}{2}$ (C) $\alpha-2$ (D) $\alpha+\frac{5}{4}$

MSI162

2. The mode of the distribution

Marks	4	5	6	7	8
No. of students	6	7	10	8	3

(A) 5 (B) 6 (C) 8 (D) 10

MSI163

3. The mean of 21 observations (all different) is 40. If each observation greater than the median are increased by 21, then mean of observations will become

(A) 50 (B) 50.5 (C) 30 (D) 45

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MSI164

4. Compute the mode for the following frequency distribution:

Size of items	0–4	4–8	8–12	12–16	16–20	20–24	24–28	28–32	32–36	36–40
Frequency	5	7	9	17	12	10	6	3	1	0

(A) 21.12 (B) 14.46 (C) 32.66 (D) 84.63

MSI165

5. If the difference between mean and mode is 63, the difference between mean and median is

(A) 189 (B) 21 (C) 31.5 (D) 48.5

MSI166

6. The range of following set of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is

(A) 11 (B) 7 (C) 5.5 (D) 6

MSI167

7. Find mean deviation from mean

x	5	7	9	10	12	15
f	8	6	2	2	2	6

(A) 3.33 (B) 3.55 (C) 3.28 (D) 3.38

MSI168

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8. The scores of a batsman in ten innings are: 38, 70, 48, 34, 42, 55, 63, 46, 54, 44. Find the mean deviation about median.

(A) $\frac{43}{5}$ (B) $\frac{44}{5}$ (C) $\frac{41}{5}$ (D) $\frac{42}{5}$

MSI169

9. If the algebraic sum of deviations of 20 observations from 30 is 20, then the mean of the observation is

(A) 30 (B) 30.1 (C) 29 (D) 31

MSI170

10. Find the mean deviation about the median of the following frequency distribution:

Class	0– 6	6– 12	12– 18	18– 24	24– 30
Frequency	8	10	12	9	5

(A) 6.318 (B) 8.218 (C) 12.910 (D) 14.218

MSI171

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ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	A	B	A	B	B	B	D	A	D	A

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RACE Regular Analysis Through Continuous Exercise

Statistics: RACE-03

Single Correct Type Questions:

1. What is standard deviation of the set of observations 32, 28, 29, 30, 31?

(A) 1.6 (B) $\sqrt{2}$ (C) 2 (D) None of these

MSI172

2. The variance of the data 2, 4, 6, 8, 10 is

(A) 6 (B) 7 (C) 8 (D) None of these

MSI173

3. The $S.D.$ of 5 scores 1, 2, 3, 4, 5 is

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

MSI174

4. If $n = 10, \overline{x} = \sqrt{12}, \sum x^2 = 1560$, then standard deviation σ is

(A) 12 (B) 13 (C) $\sqrt{166}$ (D) $\sqrt{12}$

MSI175

5. Variance of first 20 natural number is

(A) $\frac{133}{4}$ (B) $\frac{139}{12}$ (C) $\frac{133}{2}$ (D) $\frac{139}{4}$

MSI176

6. In a series of $2n$ observations, half of them equal to a and remaining half equal to $-a$. If the standard deviation of the observations is 2, then $|a|$ equals

(A) $\frac{\sqrt{2}}{n}$ (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{n}$

MSI177

7. The standard deviation of 25 numbers is 40. If each of the numbers is increased by 5, then the new standard deviation will be

(A) 40 (B) 45 (C) $40+\frac{21}{25}$ (D) None of these

MSI178

8. For $(2n+1)$ observations $x_1, -x_1, x_2, -x_2, \ldots . x_n, -x_n$ and 0 where x's are all distinct. Let $S.D.$ and $M.D.$ denote the standard deviation and median respectively. Then which of the following is always true

(A) $S.D. < M.D.$

(B) $S.D. > M.D.$

(C) $S.D. = M.D.$

(D) Nothing can be said in general about the relationship of $S.D.$ and $M.D.$

MSI179

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9. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The coefficient of variation is

(A) 50% (B) 10% (C) 40% (D) None of these

MSI180

10. The means of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2 and 6, then the other two are

(A) 2 and 9 (B) 3 and 8 (C) 4 and 7 (D) 5 and 6

MSI181

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	C	A	A	C	A	B	B	C

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RACE
Regular Analysis Through Continuous Exercise

Statistics: RACE-04

Single Correct Type Questions:

1. Calculate the standard deviation for the following data :

Wages upto (in Rs.)	15	30	45	60	75	90	105	120
No. of workers	12	30	65	107	157	202	222	230

(A) 25.883 (B) 26.883 (C) 24.833 (D) 22.338

MSI182

2. What is the standard deviation of the following series:

Measurements	0– 10	10– 20	20– 30	30– 40
Frequency	1	3	4	2

(A) 81 (B) 7.6 (C) 9 (D) 2.26

MSI183

3. Find the variance and standard deviation of the following frequency distribution:

Variable (x_i)	2	4	6	8	10	12	14	16
Frequency (f_i)	4	4	5	15	8	5	4	5

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(A) 1.38 (B) 2.38 (C) 3.38 (D) 4.38

MSI184

4. If v is the variance and σ is the standard deviation, then

(A) $v^2 = \sigma$ (B) $v = \sigma^2$ (C) $v = \frac{1}{\sigma}$ (D) $v = \frac{1}{\sigma^2}$

MSI185

5. The mean of distribution is 4 if coefficient of variation is 58%. Then standard deviation of distribution is

(A) 2.23 (B) 3.23 (C) 2.32 (D) 2.75

MSI186

6. The average weight of 9 men is $x\ kg$. After another man joins the group, the average increases by 5%. Still another man joins and average returns to old level of $x\ kg$. Which one of the following true?

(A) the 10th & 11th men weight same

(B) the 10th man weight half as much as the 11th man

(C) the 10th man weight three times the 11th man

(D) None of these

MSI187

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7. Mean of variates $1\cdot 2\cdot 3, 2\cdot 3\cdot 4, \ldots\ldots, n(n+1)(n+2)$

(A) $\frac{n(n+1)(n+2)}{4}$ (B) $\frac{(n+1)(n+2)(n+3)}{2}$

(C) $\frac{(n+1)(n+2)(n+3)}{4}$ (D) $\frac{n(n+1)(n+3)}{2}$

MSI188

8. The sum of squares of deviations for 10 observations taken from mean 50 is 250. The coefficient of variation is

(A) 50% (B) 10% (C) 40% (D) 30%

MSI189

9. The mean deviation from mean of the observations $a, a+d, a+2d, \ldots\ldots a+2^{nd}$ is

(A) $\frac{n(n+1)d^2}{3}$ (B) $\frac{n(n+1)}{2}d^2$ (C) $a+\frac{n(n+1)d^2}{2}$ (D) $\frac{n(n+1)|d|}{(2n+1)}$

MSI190

10. The marks of some students were listed out of a maximum 100. The standard deviation of marks was found to be 9. Subsequently the marks raised to a maximum of 150 and standard deviation of new marks was calculated. The new standard deviation

(A) 9 (B) 13.5 (C) –13.5 (D) –9

MSI191

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	C	B	C	C	C	B	D	B

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Important Notes

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